

Macrophages and microglia: the cerberus of glioblastoma.

PMID: 33766119 | Indexed in PubMed

Glioblastoma (GBM) is the most aggressive and deadliest of the primary brain tumors, characterized by malignant growth, invasion into the brain parenchyma, and resistance to therapy. GBM is a heterogeneous disease characterized by high degrees of both inter- and intra-tumor heterogeneity. Another layer of complexity arises from the unique brain microenvironment in which GBM develops and grows. The GBM microenvironment consists of neoplastic and non-neoplastic cells. The most abundant non-neoplastic cells are those of the innate immune system, called tumor-associated macrophages (TAMs). TAMs constitute up to 40% of the tumor mass and consist of both brain-resident microglia and bone marrow-derived myeloid cells from the periphery. Although genetically stable, TAMs can change their expression profiles based upon the signals that they receive from tumor cells; therefore, heterogeneity in GBM creates heterogeneity in TAMs. By interacting with tumor cells and with the other non-neoplastic cells in the tumor microenvironment, TAMs promote tumor progression. Here, we review the origin, heterogeneity, and functional roles of TAMs. In addition, we discuss the prospects of therapeutically targeting TAMs alone or in combination with standard or newly-emerging GBM targeting therapies.

Context-Dependent Glioblastoma-Macrophage/Microglia Symbiosis and Associated Mechanisms.

PMID: 33663953 | Indexed in PubMed

Glioblastoma (GBM) is a lethal form of primary brain tumor in human adults. The impact of tumor-intrinsic alterations is not exclusively confined to cancer cells but can also be extended to the tumor microenvironment (TME). Glioblastoma-associated macrophages/microglia (GAMs) are a prominent type of immune cells that account for up to 50% of total cells in GBM. Emerging evidence suggests that context-dependent GBM-GAM symbiotic interactions are pivotal for tumor growth and progression. Here, we discuss how specific genetic alterations in GBM cells affect GAM biology and, reciprocally, how GAMs support GBM progression. We hypothesize that understanding context-dependent GBM-GAM symbiosis may reveal the molecular basis of GBM tumorigenesis and lead to novel candidate treatment approaches aiming to improve GBM patient outcomes.

Exploiting Microglial Functions for the Treatment of Glioblastoma.

PMID: 27528361 | Indexed in PubMed

Glioblastoma multiforme (GBM) is the most common brain tumor in adults and is associated with a very low survival rate. The heterogeneity of the tumor microenvironment, its resistance to drug and radiation therapy, and its robust invasiveness all contribute to the poor outcome. Large numbers of glioma associated microglia and macrophages (GAMs) can accumulate within the tumor where they appear to have an important role in prognosis. An extensive revision of current available literature on this topic has been carried out, using the PubMed database. Articles exploring the contribution of GAMs to GBM biology as well as evidence that GAMs can be pharmacologically modulated to inhibit tumor growth are critically discussed in this review article. GAMs constitute the largest portion of tumor infiltrating cells contributing

up to 30% of the entire glioma mass. Upon interaction with neoplastic cells, GAMs acquire a unique phenotype of activation including both M1 and M2 specific markers. Different profiles of activation usually co-exist in the same tumor that is dependent upon GAM location or stage of disease. In addition to regulating immune responses which may control or favor astrocyte malignant transformation, GAMs are directly involved in the degradation of the extracellular matrix (ECM), a crucial mechanism that allows the expansion of tumors and parenchyma invasion. Several pharmacological strategies have been developed which interfere with GAM recruitment at the tumor site, cell polarization and immune function, and ECM remodeling by GAM-secreted factors. The most promising therapeutic approaches appear to target both GBM cells and GAM biological properties. GAMs significantly contribute to GBM biology (favoring tumor growth and invasiveness). Data reviewed in the present article suggest that these cells represent a valuable alternative/ additional target for the development of more effective treatments for GBM.

Immunomodulatory Effect of Microglia-Released Cytokines in Gliomas.

PMID: 33917013 | Indexed in PubMed

Microglia, a type of differentiated tissue macrophage, are considered to be the most plastic cell population of the central nervous system (CNS). Microglia substantially contribute to the growth and invasion of tumor mass in brain tumors including glioblastoma (GB). In response to pathological conditions, resting microglia undergo a stereotypic activation process and become capable of phagocytosis, antigen presentation, and lymphocyte activation. Considering their immune effector function, it is not surprising to see microglia accumulation in almost every CNS disease process, including malignant brain tumors. Large numbers of glioma associated microglia and macrophages (GAMs) can accumulate within the tumor where they appear to have an important role in prognosis. GAMs constitute the largest portion of tumor infiltrating cells, contributing up to 30% of the entire glioma mass and upon interaction with neoplastic cells. GAMs acquire a unique phenotype of activation, including both M1 and M2 specific markers. It has been demonstrated that microglia possess a dual role: on one hand, microglia may represent a CNS anti-tumor response, which is inactivated by local secretion of immunosuppressive factors by glioma cells. On the other hand, taking into account that microglia are capable of secreting a variety of immunomodulatory cytokines, it is possible that they are attracted by gliomas to promote tumor growth. A better understanding of microglia-glioma interaction will be helpful in designing novel immune-based therapies against these fatal tumors. Concluding, as microglia significantly may contribute to glioma biology, favoring tumor growth and invasiveness, these cells represent a valuable alternative/additional target for the development of more effective treatments for gliomas.

Cellular and Molecular Identity of Tumor-Associated Macrophages in Glioblastoma.

PMID: 28235764 | Indexed in PubMed

In glioblastoma (GBM), tumor-associated macrophages (TAM) represent up to one half of the cells of the tumor mass, including both infiltrating macrophages and resident brain microglia. In an effort to delineate the temporal and spatial dynamics of TAM composition during gliomagenesis, we used genetically engineered and GL261-induced mouse models in combination with CX3CR1GFP/WT;CCR2RFP/WT double knock-in mice. Using this approach, we demonstrated that CX3CR1LoCCR2Hi monocytes were recruited to the GBM, where they transitioned to CX3CR1HiCCR2Lo

macrophages and CX3CR1^{hi}CCR2⁻ microglia-like cells. Infiltrating macrophages/monocytes constituted approximately 85% of the total TAM population, with resident microglia accounting for the approximately 15% remaining. Bone marrow-derived infiltrating macrophages/monocytes were recruited to the tumor early during GBM initiation, where they localized preferentially to perivascular areas. In contrast, resident microglia were localized mainly to peritumoral regions. RNA-sequencing analyses revealed differential gene expression patterns unique to infiltrating and resident cells, suggesting unique functions for each TAM population. Notably, limiting monocyte infiltration via genetic *Ccl2* reduction prolonged the survival of tumor-bearing mice. Our findings illuminate the unique composition and functions of infiltrating and resident myeloid cells in GBM, establishing a rationale to target infiltrating cells in this neoplasm. *Cancer Res*; 77(9); 2266-78. ©2017 AACR.

Tumor-associated microglia and macrophages in glioblastoma: From basic insights to therapeutic opportunities.

PMID: 35967394 | Indexed in PubMed

Glioblastoma (GBM) is the most common and malignant primary brain tumor in adults. Currently, the standard treatment of glioblastoma includes surgery, radiotherapy, and chemotherapy. Despite aggressive treatment, the median survival is only 15 months. GBM progression and therapeutic resistance are the results of the complex interactions between tumor cells and tumor microenvironment (TME). TME consists of several different cell types, such as stromal cells, endothelial cells and immune cells. Although GBM has the immunologically "cold" characteristic with very little lymphocyte infiltration, the TME of GBM can contain more than 30% of tumor-associated microglia and macrophages (TAMs). TAMs can release cytokines and growth factors to promote tumor proliferation, survival and metastasis progression as well as inhibit the function of immune cells. Thus, TAMs are logical therapeutic targets for GBM. In this review, we discussed the characteristics and functions of the TAMs and evaluated the state of the art of TAMs-targeting strategies in GBM. This review helps to understand how TAMs promote GBM progression and summarizes the present therapeutic interventions to target TAMs. It will possibly pave the way for new immune therapeutic avenues for GBM patients.

Single-Cell Atlas Reveals Complexity of the Immunosuppressive Microenvironment of Initial and Recurrent Glioblastoma.

PMID: 32457755 | Indexed in PubMed

The Glioblastoma (GBM) immune microenvironment plays a critical role in tumor development, progression, and prognosis. A comprehensive understanding of the intricate milieu and its interactions remains unclear, and single-cell analysis is crucially needed. Leveraging mass cytometry (CyTOF), we analyzed immunocytes from 13 initial and three recurrent GBM samples and their matched peripheral blood mononuclear cells (pPBMCs). Using a panel of 30 markers, we provide a high-dimensional view of the complex GBM immune microenvironment. Hematoxylin and eosin staining and polychromatic immunofluorescence were used for verification of the key findings. In the initial and recurrent GBMs, glioma-associated microglia/macrophages (GAMs) constituted 59.05 and 27.87% of the immunocytes, respectively; programmed cell death-ligand 1 (PD-L1), T cell immunoglobulin domain and mucin domain-3 (TIM-3), lymphocyte activation gene-3 (LAG-3), interleukin-10 (IL-10) and transforming growth

factor- β (TGF β) demonstrated different expression levels in the GAMs among the patients. GAMs could be subdivided into different subgroups with different phenotypes. Both the exhausted T cell and regulatory T (Treg) cell percentages were significantly higher in tumors than in pPBMCs. The natural killer (NK) cells that infiltrated into the tumor lesions expressed higher levels of CXC chemokine receptor 3 (CXCR3), as these cells expressed lower levels of interferon- γ (IFN γ). The immune microenvironment in the initial and recurrent GBMs displayed similar suppressive changes. Our study confirmed that GAMs, as the dominant infiltrating immunocytes, present great inter- and intra-tumoral heterogeneity and that GAMs, increased exhausted T cells, infiltrating Tregs, and nonfunctional NK cells contribute to local immune suppressive characteristics. Recurrent GBMs share similar immune signatures with the initial GBMs except the proportion of GAMs decreases.

A topographic approach to the markers of macrophage/microglia and other cell types in high grade glioma.

PMID: 39734023 | Indexed in PubMed

In glioblastoma, glioma-associated microglia/macrophages (GAMs) represent the major population of tumor infiltrating cells, with up to one half of the cells of the tumor mass. Recent studies have shown that microglia are involved in the maintenance of immunological homeostasis and protection against autoimmunity. However, despite the growing body of evidence on the topic, many aspects are yet to be clarified. In our study, 3 different situations emerged concerning the markers of microglial/macrophage-related and other cell types in GBM patients: i) most of the markers (IBA1, TMEM119, CD206 and CD86) show an ascending gradient from the tumor center to the non-tumor/healthy area of the brain; ii) one marker (CD204) shows a descending gradient, going from the center of the tumor to the non-tumor/healthy brain area; iii) two markers (CD163 and P2RY12) show no gradient. These observations support the idea that the magnitude of the diverted inflammation is a 'extensive' rather than a 'local' phenomenon and that could possibly play a role in disease resistance and relapse.

Distinct regional ontogeny and activation of tumor associated macrophages in human glioblastoma.

PMID: 33177572 | Indexed in PubMed

Tumor-associated macrophages (TAMs) constitute up to 50% of tumor bulk in glioblastoma (GBM) and play an important role in tumor maintenance and progression. The recently discovered differences between invading tumour periphery and hypoxic tumor core implies that macrophage biology is also distinct by location. This may provide further insight into the observed treatment resistance to immune modulation. We hypothesize that macrophage activation occurs through processes that are distinct in tumor periphery versus core. We therefore investigated regional differences in TAM recruitment and evolution in GBM by combining open source single cell and bulk gene expression data. We used single cell gene expression data from 4 glioblastomas (total of 3589 cells) and 122 total bulk samples obtained from 10 different patients. Cell identity, ontogeny (bone-marrow derived macrophages-BMDM vs microglia), and macrophage activation state were inferred using verified gene expression signatures. We captured the spectrum of immune states using cell trajectory analysis with pseudotime ordering. In

keeping with previous studies, TAMs carrying BMDM identity were more abundant in tumor bulk while microglia-derived TAMs dominated the tumor periphery across all macrophage activation states including pre-activation. We note that core TAMs evolve towards a pro-inflammatory state and identify a subpopulation of cells based on a gene program exhibiting strong, opposing correlation with Programmed cell Death-1 (PD-1) signaling, which may correlate to their response to PD-1 inhibition. By contrast, peripheral TAMs evolve towards anti-inflammatory phenotype and contains a population of cells strongly associated with NFkB signaling. Our preliminary analysis suggests important regional differences in TAMs with regard to recruitment and evolution. We identify regionally distinct and potentially actionable cell subpopulations and advocate the need for a multi-targeted approach to GBM therapeutics.

A Heterotypic Tridimensional Model to Study the Interaction of Macrophages and Glioblastoma In Vitro.

PMID: 34065977 | Indexed in PubMed

Glioblastoma multiforme (GBM) is the most frequent and aggressive primary brain tumor, and macrophages account for 30-40% of its composition. Most of these macrophages derive from bone marrow monocytes playing a crucial role in tumor progression. Unraveling the mechanisms of macrophages-GBM crosstalk in an appropriate model will contribute to the development of specific and more successful therapies. We investigated the interaction of U87MG human GBM cells with primary human CD14+ monocytes or the THP-1 cell line with the aim of establishing a physiologically relevant heterotypic culture model. primary monocytes and THP-1 cells were cultured in the presence of U87MG conditioned media or co-cultured together with previously formed GBM spheroids. Monocyte differentiation was determined by flow cytometry. primary monocytes differentiate to M2 macrophages when incubated with U87MG conditioned media in 2-dimensional culture, as determined by the increased percentage of CD14+CD206+ and CD64+CD206+ populations in CD11b+ cells. Moreover, the mitochondrial protein p32/gC1qR is expressed in monocytes exposed to U87MG conditioned media. When primary CD14+ monocytes or THP-1 cells are added to previously formed GBM spheroids, both invade and establish within them. However, only primary monocytes differentiate and acquire a clear M2 phenotype characterized by the upregulation of CD206, CD163, and MERTK surface markers on the CD11b+CD14+ population and induce alterations in the sphericity of the cell cultures. our results present a new physiologically relevant model to study GBM/macrophage interactions in a human setting and suggest that both soluble GBM factors, as well as cell-contact dependent signals, are strong inducers of anti-inflammatory macrophages within the tumor niche.
